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Annealing and intensity dependent femtosecond nonlinear optical studies of sodium zinc borate oxide glasses incorporated with gold nanoparticles

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Highlights

- Influence of annealing on optical nonlinear features gold nanoparticles embedded sodium zinc borate glasses has been studied.
- Nonlinear optical traits have been evaluated under femtosecond pulse regime.
- The intensity-sensitive nonlinear optical features were evaluated.
- The results endorse gold nanoparticles embedded sodium zinc borate glass is beneficial for nonlinear photonic applications.

Abstract

Herein, we explored the annealing and intensity-dependent femtosecond (fs) nonlinear optical (NLO) studies on sodium zinc borate (NZB) glasses embedded with gold nanoparticles (NPs). The glasses were prepared by the melt-quench process and subjected for thermal treatment at 450°C at different schedules in a normal atmosphere. It was identified that the glass annealed for 12 hr (i.e., the NZBAu12 glass) had the maximum saturation intensity and negative nonlinear refraction values. Therefore, the intensity-sensitive NLO investigations then conducted on that glass to understand the influence of intensity of laser radiation on NLO properties on gold NPs embedded glasses. The saturable absorption observed at lower intensity was switched to reverse saturable absorption (two-photon absorption) during the exposure with high fluence laser pulses. However, the nonlinear refraction sign was unchanged with increase of intensity of laser radiation. The strategy explored in this work might pave the way for achieving the strong nonlinear traits in borate-based glasses routed with gold NPs. The study outcomes endorse the benefits of NZBAu12 glass for optical switching functionalities under low-intensity pulse excitation, as well as for optical limiting applications under high-intensity pulse excitation.

Keywords

Borate glasses; Gold nanoparticles; Nonlinear optical; Annealing dependent; Intensity dependent; Near-infrared region

CRediT authorship contribution statement

Declaration of Competing Interest

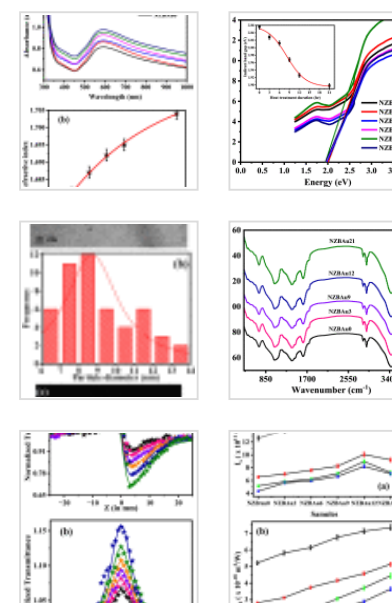
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1. Introduction

When the bulk gold is reduced to nanoscale, this results in interesting features to be used in various functional applications. Au nanoparticles (NPs) absorb and scatter light efficiently due to their significant interactions with radiation. This happens when the conduction electrons located on the metal surface collectively oscillate, with the activation by light at certain wavelengths [1]. The absorption and scattering intensities of Au NPs are significantly higher than similarly sized non-plasmonic NPs due to an oscillation known as a surface plasmon resonance (SPR) [2]. Owing to the strong interaction with matter and fascinating properties of gold NPs, these NPs have been explored for diverse functionalities, including cancer diagnosis, photothermal therapy [3], super capacitor [4], photonic [5], and surface-enhanced Raman scattering [6] applications. Additionally, the Au NPs have been explored for optical limiting [7], optical switching [8], and harmonic generation [9] nonlinear optical (NLO) functionalities in the recent decades. Further, gold NPs' NLO features are characterized by sensitivity to their size, shape and concentrations [10], [11], [12]. NLO properties are well known to be widely employed in designing advanced technologies ranging from signal processing, telecommunications, super-resolution lithography, data storage, and microscopy to terahertz and higher harmonic generation, fiber optics applications, and also space components. In particular, the NLO materials are crucial for developing next-generation integrated photonic circuits [13].

It has been documented that when the Au NPs routed to a host of the materials causes strong enhancement of NLO features through local field, resonance and/or energy transfer mechanisms [10,[14], [15], [16]]. However, most of the investigations devoted to embedding the gold NPs to one dimension and two-dimensional media. The difficulty of these one- and two-dimensional hosts preparation in different shapes and sizes hinders their practical use in the functional applications. To this end, inorganic glasses have received tremendous interest since the glasses can be fabricated in desired shapes and sizes including three- dimensions enabling them to use in practical applications effectively [17]. Besides, the glasses possess high laser and mechanical damage thresholds. Therefore, the glass hosts can be utilized to execute the functioning under

high laser fluences of various experimental conditions [18,19]. However, the glasses possess weak NLO features compared to many advanced one- and two-dimensional hosts. It has been well documented that glass specimens' NLO attributes can be considerably enhanced by loading the metal NPs such as gold [12], silver, and copper [20] NPs by melt quenching [21], ion implantation [22], and ion-exchange [23] processes. Further, the borate glasses can withstand higher fluences of laser light without any host damage, which is important for the devices to execute the functioning for longer life and without any reduction in the efficiency. Therefore, the borate-based glass hosts are believed to be of high interest among different glass families for NLO applications.

Generally, for optoelectronic and photonic applications the materials with strong NLO absorptive and refractive coefficients are beneficial [24]. While, the hosts with large NLO absorption and weak nonlinear refractive index (n_2) along with low optical limiting threshold are important for optical limiting applications [25]. Alternatively, the small linear and nonlinear absorption coefficient, strong n_2 along with fast response times materials are essential for optical switching devices [26]. Additionally, the materials exhibiting saturable absorption (SA) are more preferable for short pulse generation (passive Q-switching) applications [27]. With aim of developing the glasses for optical switching applications, we prepared and carried out the NLO investigations on the gold NP-embedded sodium zinc borate glasses in our previous study [28], in which the concentration of single-valent alkali oxide was varied from 10 to 30 mol% to tune the NLO features. It was emphasized that the saturation intensity (I_s , the essential parameter to be considered in designating high power lasers delivering ultrashort laser pulses) and n_2 are optimized (highest) in 30% Na₂O containing glasses. Usually, the properties of metal NP-embedded glasses/materials can be improved by varying the metal NPs quantity in the composition [12], and by applying heat treatment to the optimized glass system at certain durations and/or temperatures [5,29]. Therefore, in order to tune the I_s and n_2 values of sodium zinc borate glasses embedded with gold NPs, the optimized glass composition was prepared and annealed at 450°C for diverse durations. Finally, the investigation of

intensity-dependent NLO was conducted on the Au NP-embedded sodium zinc borate glass that possess highest NLO features.

2. Materials and methods

The glass bearing molar composition, $30\text{Na}_2\text{O}-10\text{ZnO}-60\text{B}_2\text{O}_3-0.1\text{AuCl}_3$ was fabricated through the melt quenching process. Laboratory reagent grade chemicals such as sodium carbonate, zinc oxide, boric and chloroauric acids of 99.99% were purchased from SD Fine Chem. Ltd. and utilized as starting precursors for synthesizing glass specimens. Fine grinding of all the precursors for a 30 gm batch was carried out, after which they were placed into a porcelain crucible. The glass batch's sodium carbonate was decarbonized at 550°C for around 20min, with the temperature then increased to 1050°C , at which point the mixture was completely melted. Finally, after 30min at melting temperature with constant stirring at 10min intervals, the glass melt was quenched to room temperature. Annealing of the as-prepared glass for 4 hr at 300°C was carried out to eliminate the internal stresses that manifested while quenching to room temperature from melting temperature. The glass was cut in to small circularly shaped discs of diameter 10mm and then subjected for heat treatment at 450°C at 0, 3, 6, 9, 12 and 21 hr. Mirror polishing of the heat-treated glass specimens was then conducted for optical and NLO characterizations, after which they were finely crushed into powders for Fourier transform infrared (FTIR) and microscopic characterizations. The NZBAux label was applied to the glass specimens post-thermal treatment at 450°C , where 'x' instantiates the period of thermal treatment.

The Perkin Elmer Lambda 35 UV Vis spectrophotometer was used to measure the glass specimens' optical absorption spectral outcomes in the region of 200–1100nm ($\pm 2\text{nm}$). The refractive index (n) magnitudes of the glass specimens were measured utilizing Abbe's refractometer in which the sodium light (wavelength for D line: 589.3nm) was benighted as the radiation source. In each sample set, the 'n' value was measured on three different samples and averaged values are presented here. A Thermo Nicolet Avatar 360 FTIR spectrometer was employed to collect the glass powders' FTIR spectral outcomes in

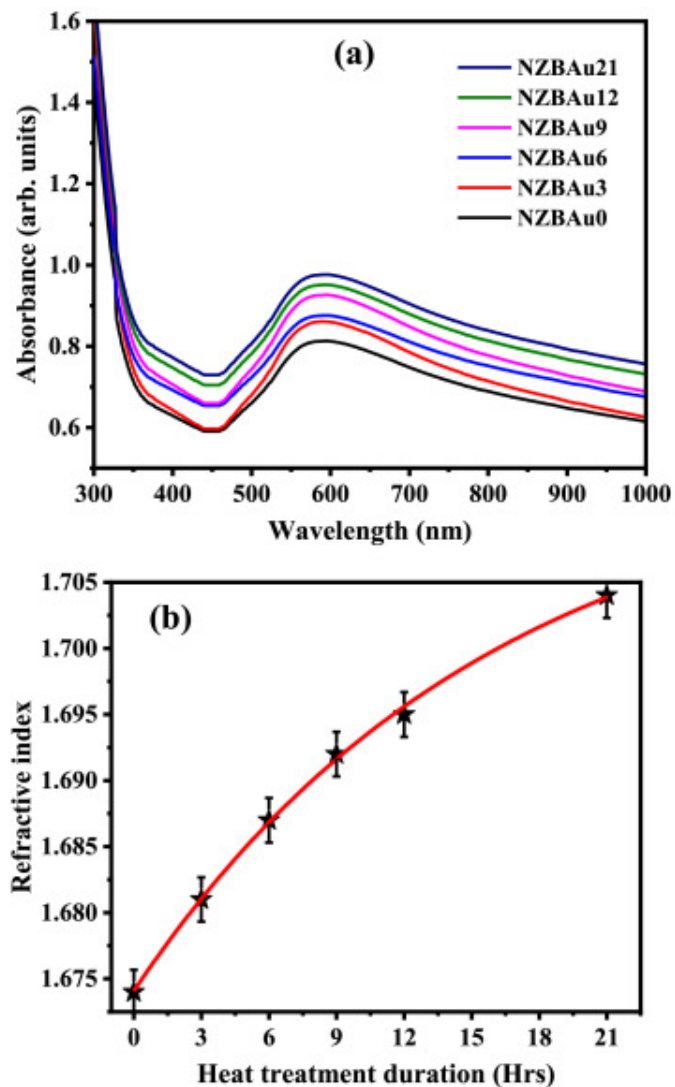
the 400–4000 cm^{-1} wavenumber range ($\pm 4 \text{ cm}^{-1}$) with a 4 cm^{-1} interval using the KBr pellet method. High-resolution transmission electron microscopy (HR-TEM) measurements were conducted utilizing a Tecnai G2 F30. Assessment of the NLO properties (e.g., NLO absorption and NLO refractions) was through the open aperture (OA) and closed aperture (CA) Z-scan configurations. The Z-scan analysis was conducted from 700nm to 1000nm (in steps of 100nm) under the femtosecond (fs) pulse regime using 150 fs laser pulses from a Chameleon Ti:Sapphire laser (Coherent Inc., USA) at 80MHz repetition rate. The detailed procedure of Z-scan experiment executed in fs regime can be found elsewhere [12,[26], [30], [31]]. Execution of the OA and CA Z-scan measurements were at five parts (spots) of each sample, with the average results presented here. It was identified that the uncertainties in the estimation of NLO absorption and refraction coefficients are found to be 3% and 5%, respectively, that might be caused due to poor polishing quality of the sample and fluctuations in the input laser.

3. Results and discussion

3.1. Optical and structural properties

The NZBAux glasses' linear optical absorption spectra are displayed in Fig.1(a) in the 300–1000nm spectral region. A broad and robust band in the 462–757nm (for NZBAu0 glass) visible spectral region is clearly revealed in the spectra exhibited in Fig.1(a), which corresponds to the Au^0 NPs SPR that arose due to the collective oscillation electrons found at the two materials' interface (e.g., metal–dielectric), where the real part of the dielectric function switches sign through the interface expense [32]. It can be noticed from Fig.1(a) that the SPR peak of the Au NPs' central maxima has moved in the direction of the longer wavelengths with an increase in annealing duration. The SPR maxima's red-shift of Au^0 NPs embedded in the glass host seen in the current study is manifested as Au^0 NP growth through thermal treatment of a longer period [33]. Usually, the NPs growth occurs through Ostwald ripening, and NPs movement and agglomeration processes [34]. Further, it is believed that during the glasses' preparation, there was a dissociation of the

included AuCl_3 into gold and Cl^- ions, and most of gold was dissolved in the glass composition in the form of ionic species and some fraction of the gold gets dissolved in the form of Au^0 . When subjecting the as-prepared glass to annealing, there was a reduction of the dissolved ionic gold to Au^0 with the formation of NPs. Increasing amounts of ionic Au were reduced to neutral Au with increased annealing duration, thus leading to increased Au NP concentration. Hence, due to the reduced Au^0 NP fraction in the NZBAu0 glass, there was less peak intensity of the SPR peak, while, as a result of major ionic-Au-to-neutral-Au fraction conversion with annealing the increased SPR peak intensity at longer durations of annealed glasses is evidenced from optical absorption spectral results [12,35].



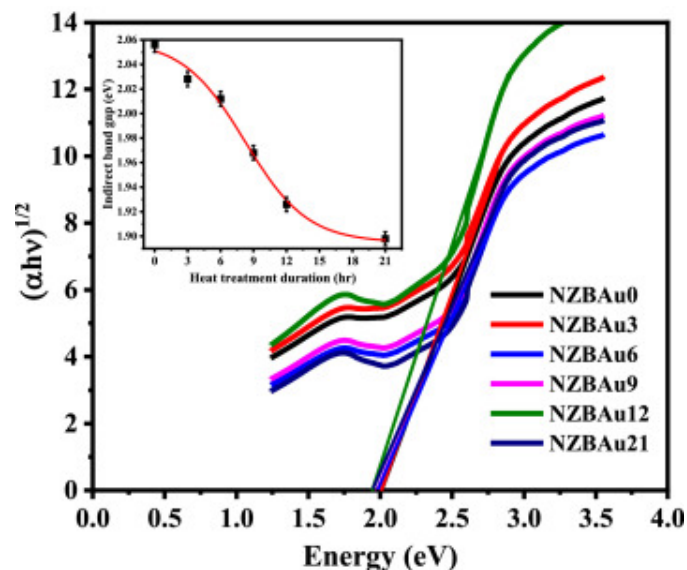
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Fig. 1. (a) The studied glasses' optical absorption spectral outcomes, (b) the variation of 'n' of all the NZBAux glass as a function of the annealing period.

[Fig. 1\(b\)](#) depicts the change of 'n' as an annealing duration function, highlighting a monotonous enhancement of the studied glass host's 'n' with the thermal schedule. It is largely known that, the 'n' of the host and concentration-size of the embedded NPs are directly interrelated [12,36]. Hence, there is a corroboration of the monotonous elevation of 'n' through the annealing schedule to the concentration and size of Au⁰ NPs routed in sodium zinc borate glass. Additionally, the SPR maxima's red-shift as seen in optical absorption results is also manifested for this increase of 'n' of annealed glasses.

Since the indirect bandgap (E_g) is believed to occur in disordered materials, the Tauc's plots were examined to compute the E_g factors [37] for the studied glass samples, and the data is displayed in [Fig. 2](#). In terms of the annealing duration, the variation of the studied glasses' E_g values is depicted in the [Fig. 2](#) inset, from which it can be evidenced that the E_g magnitude of glasses found to reduces with respect to thermal treatment. This reduce of E_g of glass system with thermal schedule is manifested to increase of disorder due to the glass's reduced rigidity that resulted with annealing duration. When the glass host routed with NPs are subjected to annealing, the growth of NPs was believed to usually occur due to the migration of NPs followed by coalesce [38]. On account of this growth of Au NPs in the studied glasses with respect to annealing duration, the non-bridging oxygens (NBOs) could be formed within the glass structure as the NPs' growth in glass network has a disrupting effect on the bridging bonds. This generation of NBOs resulted in the enactment of disorder of the glass that ultimately reduces the glass matrix's rigidity, leading to a reduction in the glass's E_g [38].



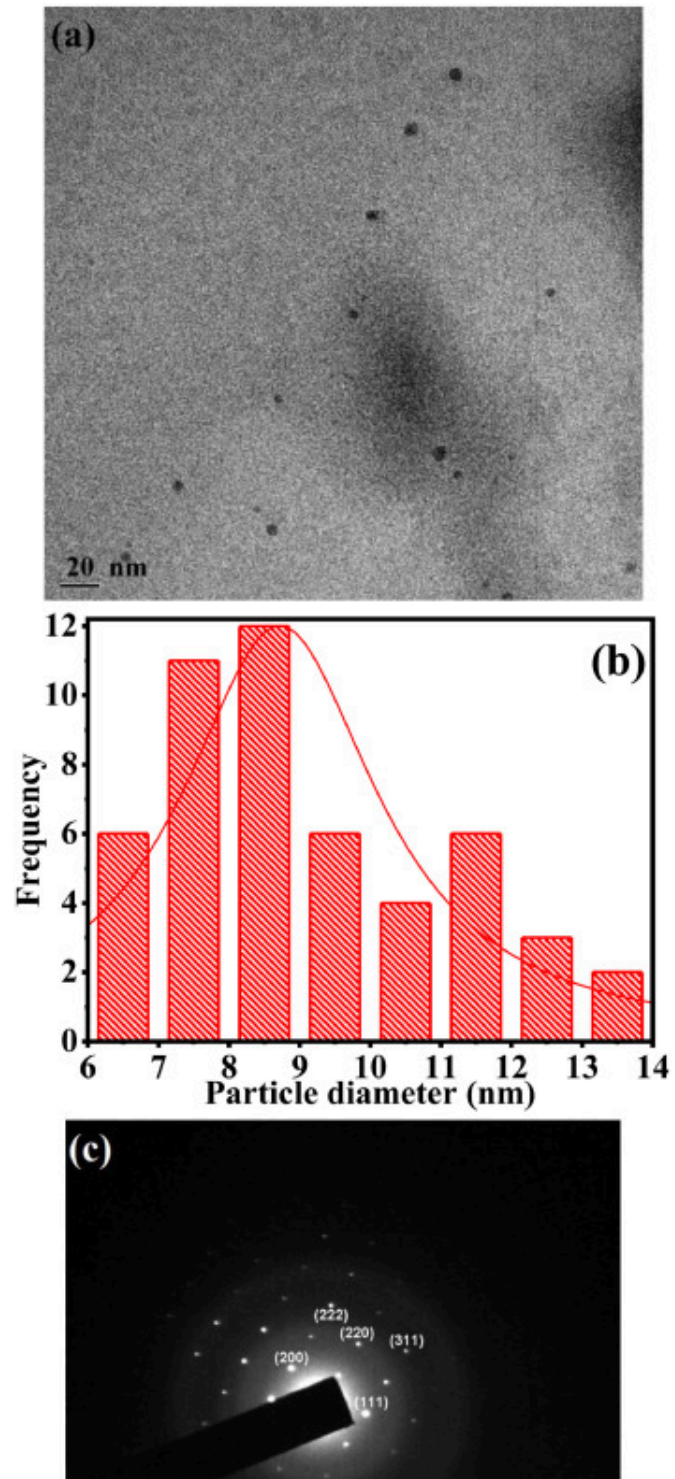
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Fig. 2. Tauc's plots for all the studied glasses used in the current work for evaluating the E_g values. Inset illustrates the variation of E_g values regarding the annealing schedule.

The morphology of the incorporated Au NPs has been analysed utilizing HR-TEM characterization, with the image of the NZBAu3 specimen displayed Fig.3(a). The micrograph in Fig.3(a) represents Au NPs' homogeneous dispersion in the studied glass host, where it is evident that most of the embedded Au NPs have a spherical form. From the micrograph, it is evident that the glass also consists of minor content of Au NPs that have a non-spherical form. The characteristics such as mean diameter and size distribution of the Au NPs loaded in NZBAu3 glass were analysed by plotting the histogram, with Fig.3(b) depicting that the embedded Au NPs ranged from 6.27 to 11.98 nm in size, with a median size of 8.64 nm. Further, the NPs' average size was theoretically determined by employing the Mie theory to the optical absorption spectrum of NZBAu3 and found to be ~ 8.46 nm. Here, the median size of Au NPs embedded in NZBAu3 glass estimated experimentally corroborated well with the theoretically

computed average size. The theoretically evaluated median sizes of the Au NPs in NZBAu6, NZBAu9, NZBAu12 and NZBAu21 glass were found to be 11.41 (± 0.05) nm, 12.31 (± 0.05) nm, 14.06 (± 0.05) nm and 19.12 (± 0.05) nm, respectively. From this theoretical calculation of the Au NPs' size based on Mie theory, an increase can be inferred with annealing duration. Therefore, the Au NPs' SPR peak red-shift and enhancement of 'n' with annealing schedule as mentioned in previous paragraphs is related to the NPs' growth with increased duration of thermal treatment duration [33]. The selected area electron diffraction (SAED) recorded for NZABAu3 glass is presented in Fig.3(c). The spherical rings surrounding the central bright region formed in the SAED pattern confirmed the formation of Au⁰ NPs in the investigated glass, with the pattern also revealing that the Au⁰ NP planes correspond to the (111), (200), (220), (222) and (311) planes of face-centered cubic structure of gold NPs (JCPDS card no. 4-784).





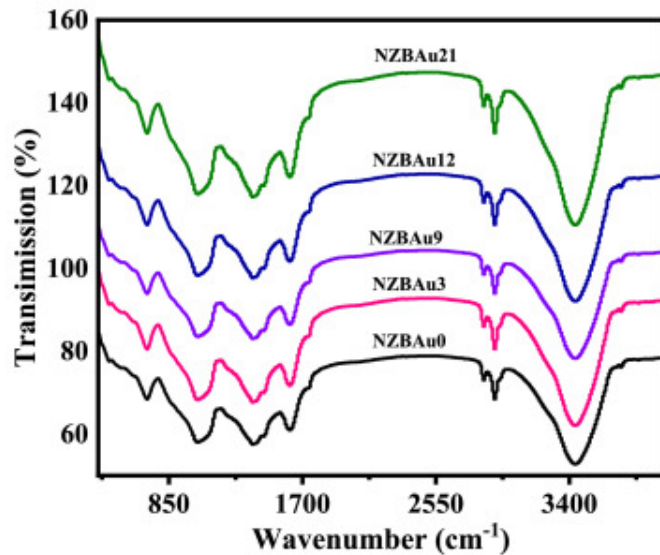
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Fig. 3. (a) HR-TEM micrograph of NZBAu3 glass, (b) corresponding histogram obtained using ImageJ software, and (c) SAED pattern of Au⁰ NPs in NZBAu3 glass.

Fig.4 depicts the studied glass samples' FTIR results. Usually, the FTIR spectral bands observed for glasses manifest in the coalescence of different vibrational levels of lattice dispersion, thermal widening, mechanical scattering, and strong degeneracy of the powdered samples. In general, the borate structure produces the FTIR bands in three regions such as 1150–1650 cm⁻¹, 750–1150 cm⁻¹ and around 700 cm⁻¹ owing to asymmetric stretching motion of B–O linkages of BO₃ groups, stretching vibrational motion of B–O linkages belonging to tetrahedral BO₄ groups and bending motion of B–O–B linkages, respectively [39]. The FTIR band observed at 1620 cm⁻¹ in Fig.4 is attributed to bending vibration of O–H connections entered to the glass during the preparation of glass. The FTIR band noted at ~ 1400 cm⁻¹ is caused due to the B–O bonds' symmetric stretching motion, which are related trigonal BO₃ groups, while the band evidenced at ~ 1050 cm⁻¹ is due to the B–O bonds' stretching motion, which are part of BO₄ tetrahedral groups [40], whereas the weak band noticed at ~ 855 cm⁻¹ corresponds to stretching vibrations of B–O units belonging to BO₄ groups from diborate units [39] and the weak band at ~ 706 cm⁻¹ originated due to the bending motion of B–O–B connections existing in the borate glass structure [41]. Further, the weaker band noticed in FTIR spectra of tested glasses positioned at ~ 464 cm⁻¹ is corroborated to the vibration of ZnO₄ network units exist in the structure of studied glasses [41]. Furthermore, the non-appearance of the band at ~ 806 cm⁻¹ in the tested glass specimens' data suggests the lack of boroxol rings in the sodium zinc borate glass system structure considered in the present

investigation [42]. The spectra appended in Fig.4 are almost identical to one another and no any new bands have been observed in longer annealed glass which instantiate the annealing schedule performed for the current study has not caused significant modifications in the structure of glass and only resulted in increase of the Au⁰ atoms / Au NPs concentration along with the growth.



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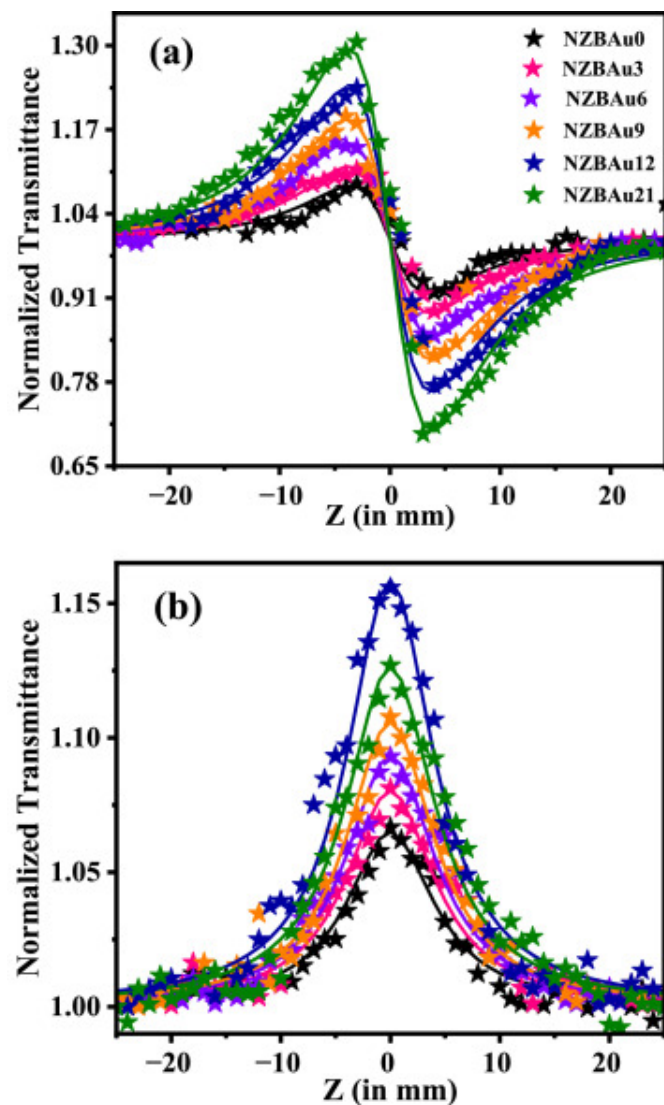
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Fig. 4. The studied glasses' FTIR spectral outcomes.

3.2. Annealing and intensity dependent NLO traits

Fig.5(a) and Fig.5(b) respectively depict the CA and OA Z-scan signatures of NZBAux glasses at the 900nm excitation wavelength. As shown in Fig.5(b), the CA Z-scan results revealed that the output transmittance attained maximum in pre-focal region followed by a minimum in post-focal region when the sample moved from the one end of the far field to the other end of far field through the focal point. This peak followed by the valley kind of signature seen in the CA Z-scan indicates the presence of negative nonlinear

refraction ($n_2 < 0$) [11] in the investigated gold NP-embedded NZB glasses. This negative nonlinear refraction occurs as a result of the self-defocusing effect of high fluence laser light irradiation [11]. The results of the OA Z-scan revealed an increase in the output transmittance when the sample shifted close to the focus with the maximum at the focal point of the scan, followed by a decrease as the sample shifted away from the focus as depicted in Fig.5(b) for the 900nm laser light exposing. This peak kind of behavior observed in OA Z-scan measurements is the characteristic of SA type of nonlinear absorption existing in the gold NP-embedded glasses [43]. At all the other excitations, both the OA and CA Z-scan signatures were identical as evidenced at 900nm excitation, which indicated that no change of NLO absorption and refraction phenomena occurred while the excitation wavelength varied.

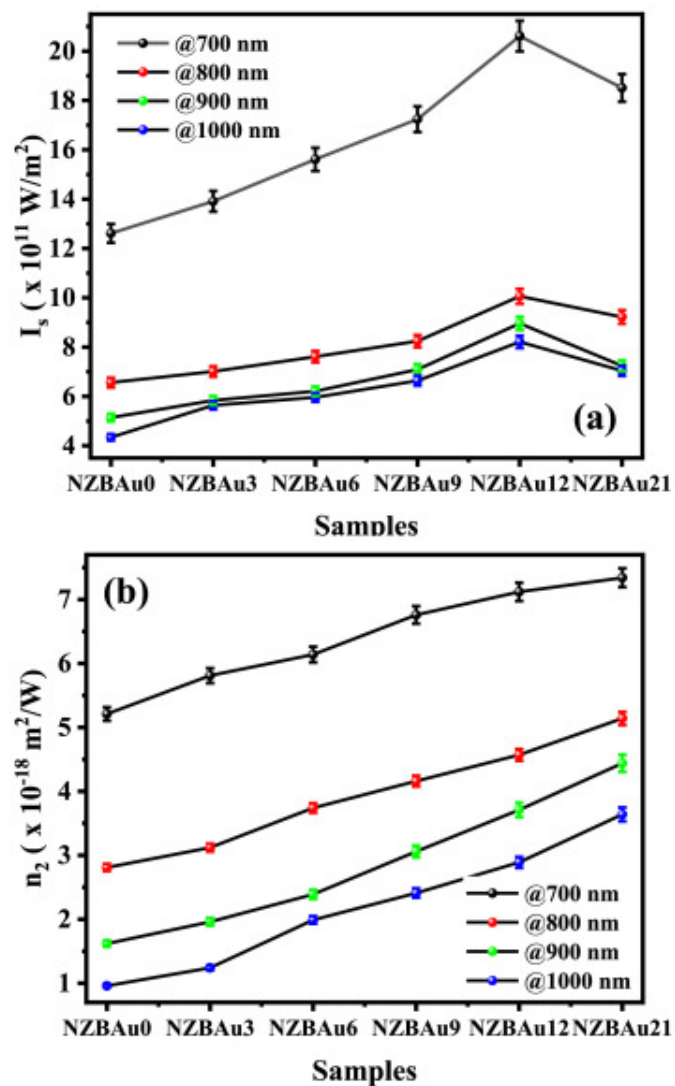


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Fig. 5. (a) The CA and (b) OA Z-scan results of the glass samples collected at 900nm irradiation at an intensity of $59.8 \times 10^{10} \text{ W/m}^2$ (solid symbols: collected data points; thick lines: theoretical data).

The theoretical fittings to the experimental data were performed on all the data by following the theoretical equations described in the earlier works [12,30,35]. The computed the I_S , $-n_2$, and two photon absorption coefficient (α_2) magnitudes for all the glass samples at all the irradiated wavelengths and at all the intensities of the laser beam. Fig.6(a) and Fig.6(b) respectively depict the variation of I_S and n_2 values of NZBAux glasses with respect to annealing duration for all the irradiated wavelengths. It can be seen from Fig.6(a) that the I_S increases with thermal treatment schedule up to 12 hr, beyond which (or at 21 hr) the I_S found to deteriorate, while Fig.6(b) shows a monotonous enhance of $-n_2$ with annealing time.



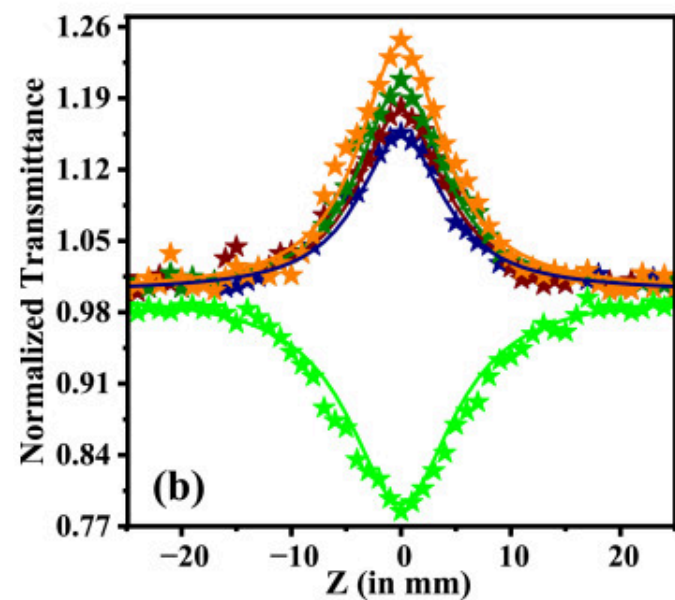
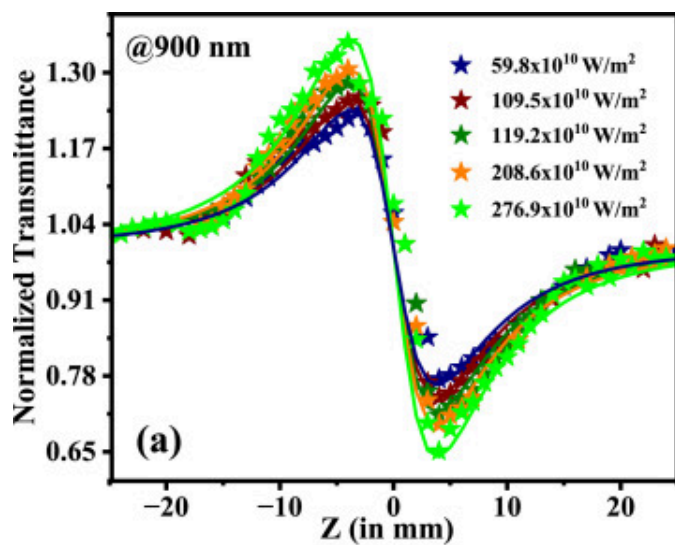
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Fig. 6. The variation of (a) I_s and (b) $|n_2|$ values of investigated Au NP-embedded sodium zinc borate glass host as the thermal treatment function at all pumped wavelengths.

Intensity dependent CA and OA Z-scan measurements were also carried out on the NZBAu12 glass (optimized/ superior compared to all the other investigated glass samples) to understand influence of increase of the input intensity of laser light on NLO features. Fig. 7(a) presents the results of the CA Z-scan at the 900nm irradiation for diverse input intensities, which did not reveal any change of nonlinear refraction with variation of input intensity of the radiation. Only, the height / distance of the peak-valley rose with increased laser radiation fluence. Fig. 7(b) provides the typical OA Z-scan outcomes recorded at the pumping of 900nm irradiation for certain input intensities. It can be seen that at the initial laser light intensity, the OA Z-scan signature has not undergone any change, although with further increase of intensity of input light, the signature reversed (i.e., peak to valley at the scan focus). This decrease of output transmittance with the sample's movement toward the focus and minimum attainment at the scan's focal point is related to the reverse saturable absorption (RSA) kind of nonlinearity [44]. This clearly revealed that the SA realized under lower intensity switched to RSA under higher input intensities in NZBAu12 glass. This RSA nonlinearity noticed in the NZBAu12 glass samples for higher input intensities is possible due to simultaneous absorption of more than one photon (two or three and so on) of laser light by the glass samples during the exposure [30]. In theoretical fitting of RSA data obtained at higher intensity of the laser beam, the I_s was considered to be zero as no SA was observed at higher input intensity of the laser beam. Therefore, only α_2 was computed at higher intensities of laser radiation for RSA data. The intensity dependent NLO characteristics retrieved for NZBAu12 glass are furnished in Table 1. In the design of optical switching devices, high I_s and strong n_2 are vital parameters, while strong α_2 are essential for data storage and optical limiting applications. Therefore, the high I_s and n_2 values achieved for NZBAu12 glass suggest that this glass is useful for optical switching functionalities to work under medium/low intensity of the laser pulses, while also being applicable for optical limiting and additional optoelectronic uses at higher intensities. Table 2 presents the NLO coefficients of different glass systems embedded with gold and silver NPs, as well as the highest values obtained for NZBAu12 glass. It is clear that highest NLO traits obtained for NBA12 are comparable with that of silver and gold NP-embedded borate and phosphate-based glasses. However,

the values are lower compared to those obtained for silver NP-embedded heavy metal borate/heavy metal oxide glasses.



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Fig. 7. (a) The CA and (b) OA Z-scan signatures obtained at 900nm irradiation at differing peak intensities in the 59.8×10^{10} to $276.9 \times 10^{10} \text{ W/m}^2$ range (solid lines illustrate the theoretical fittings to the experimental data points represented by solid symbols).

Table 1. NZBAu12 glass's intensity-sensitive NLO traits at diverse pump wavelengths.

Input Intensity ($\times 10^{10} \text{ W/m}^2$)	NLO parameters		
	α_2 ($\times 10^{11} \text{ m/W}$) ($\pm 3\%$)	I_s ($\times 10^{11} \text{ W/m}^2$) ($\pm 3\%$)	$ n_2 $ ($\times 10^{-18} \text{ m}^2/\text{W}$) ($\pm 5\%$)
$\lambda = 700 \text{ nm}$			
98	0.004	20.61	7.12
158	0.003	21.42	7.31
311	0.005	24.31	7.64
407	0.006	27.61	7.82
514	1.72	–	8.02
$\lambda = 800 \text{ nm}$			
77.6	0.001	10.06	4.57
153.4	0.003	10.71	4.82
215.8	0.004	11.39	5.03
306.9	0.005	12.69	5.21
399.4	1.43	–	5.33
$\lambda = 900 \text{ nm}$			
59.8	0.004	8.96	3.71

Input Intensity ($\times 10^{10} \text{W/m}^2$)	NLO parameters		
	α_2 ($\times 10^{11} \text{m/W}$) ($\pm 3\%$)	I_s ($\times 10^{11} \text{W/m}^2$) ($\pm 3\%$)	$ n_2 $ ($\times 10^{-18} \text{m}^2/\text{W}$) ($\pm 5\%$)
109.5	0.003	9.22	3.96
119.2	0.005	9.75	4.19
208.6	0.007	10.21	4.36
276.9	1.19	–	4.82
	$\lambda = 1000 \text{ nm}$		
49.6	0.004	8.21	2.89
99.7	0.006	8.73	3.17
127.6	0.006	9.12	3.48
199.8	0.002	9.79	3.77
236.7	0.94	–	4.09

Table 2. NLO coefficients of certain recently reported gold and silver NP-embedded glasses along with the highest values found in the current study.

Glass sample details	Excitation wavelength and other details	I_s (W/m^2)	$ n_2 $ (m^2/W)	α_2 (m/W)	Reference
20Na ₂ O–10ZnO–70B ₂ O ₃ ion-exchanged with HAuCl ₄	700nm (150 fs, 80MHz)	–	0.72×10^{-18}	0.78×10^{-11}	[23]

Glass sample details	Excitation wavelength and other details	I_s (W/m ²)	$ n_2 $ (m ² /W)	α_2 (m/W)	Reference
70SiO ₂ –20Na ₂ O–10CaO–1Sb ₂ O ₃ –0.1AuCl	813 nm (35 fs, 1 KHz)	–	1.85×10^{-20}	3.76×10^{-15}	[52]
(70Pb ₂ P ₂ O ₇ –30WO ₃):0.5CuO annealed at 410°C for 2 hr	775 nm (150 fs, 1KHz)	–	2.0×10^{-19}	–	[53]
10La ₂ O ₃ –30Pbo–60B ₂ O ₃ doped with 0.002 mol% of AuCl ₃	532 nm (5 ns, 10Hz)	18×10^{12}	–	1.89×10^{-11}	[54]
29.9Na ₂ O–10ZnO–60B ₂ O ₃ –0.1AuCl ₃	850 nm (150 fs, 80MHz)	3.9×10^{-10}	-2.6×10^{-17}	7.2×10^{-14}	[28]
95(Ge ₂ O ₃ Sn ₅ Se ₇₅)–5Au	3000 nm	–	9.7×10^{-18}	–	[55]
20Li ₂ O–15CaF ₂ –20ZnO–45B ₂ O ₃ –0.1AuCl ₃ (wt%)	800 nm (150 fs, 80MHz)	–	0.42×10^{-18}	1.64×10^{-11}	[56]
40Bi ₂ O ₃ –50B ₂ O ₃ –10SiO ₂	800 nm (200 fs, 76MHz)	5.1×10^{11}	11.8×10^{-18}	–	[57]
60Bi ₂ O ₃ –30B ₂ O ₃ –10SiO ₂ (mol%) doped with 5 wt% of AgCl	800 nm (200 fs, 76MHz)	–	54.8×10^{-18}	1.9×10^{-11}	[58]
NZBAu12	700 nm (150 fs, 80MHz)	18.51×10^{11}	7.34×10^{-18}	–	Present work

When gold NPs are exposed to electromagnetic radiation, three transitions generally occur [[14], [45]]: (i) at 670 nm, owing to an intraband (sp→sp) HOMO to LUMO

transition; (ii) at 450nm, due to mixed intraband ($sp \rightarrow sp$) and interband ($d \rightarrow sp$) transitions; and (iii) at 400nm, caused by interband transition ($d \rightarrow sp$). It is well known that intraband transitions are plausible with low/medium energy photons, whilst the interband transitions occur in the presence of higher energy photons. In the NLO measurements of the current work, the Au NP-embedded glasses were excited with low-energy near-infrared light photons. Moreover, in the plasmonic Au NP-embedded glasses, there will be a prominent Au NP SPR response [14]. Owing to the SPR dominant role in the current study the SA nonlinearity has resulted [14,46] as seen from the OA Z-scan measurements of NZBAux glasses. Further, the Au NPs' luminescence efficiency is ultra-low, thereby the Au NPs' SPR decay through intraband transitions is expected to be ultrafast [47]. Thus, there is rapid electron depletion in the vicinity of the Fermi surface through SPR, which produces the SA nonlinearity [47]. Usually, the SA (and also RSA) of nanomaterials is dependent on size and shape of the nanoparticles. It has been highlighted that the NLO absorption traits improves up to certain size threshold, beyond which the NLO traits decrease or change their sign [33,48,12]. Fig.6(a) reveals that the I_S increases with thermal treatment duration up to 12 hr, beyond which (at 21 hr) the I_S was found to deteriorate. Since the embedded Au NPs' size increases with increased heat treatment, the ' I_S ' is found to increase. However, it is believed that in the 21-hr heat treated sample the size of Au NPs might be equivalent to threshold size, at which the interaction between the substrate and NPs dominated and possibly resulted in inhomogeneous displacement of the electronic cloud. This generates the higher order multipolar charge distributions and, therefore, the I_S value decreased. Generally, with increased laser radiation intensity, the rate of depletion of electrons in the ground state of Au NPs through intraband transitions increases, and thereby the I_S value NZBAu12 glass increases. However, at very high laser radiation intensities, in addition to intraband transitions, interband and intraband-intraband transitions could occur, with the conduction band producing free carriers that absorb more photons from the high fluence laser pulses [14,49,50], which result in the strong RSA nonlinearity. Therefore, the SA has been switched to RSA in NZBAu12 glass at high intensity irradiation. Overall, it is believed that at low intensity of the laser light, the SPR bleaching through intraband transitions

dominates, which has resulted in a SA kind of nonlinearity. Whilst at higher laser radiation intensity, as well as intraband transitions, interband and mixed intraband–interband transitions occur, which in the excited state produces free carriers. Therefore, the free carrier absorption dominates over the SPR bleaching at higher fluence that causes the occurrence of RSA due to a 2PA process.

The SPR of NPs embedded in glasses stimulates the local field during the high–energy laser pulse excitation, which enhances the glass nanocomposite's n_2 [33,51]. With increase of thermal treatment schedule the size of the NPs embedded in glass increases with decrease of their interparticle spacing [33], and it obviously has resulted in the significant enhancement of local field. This local field is also proportionally enhanced with intensity of laser radiation. Thereby, the n_2 of the glass nanocomposites increases with thermal treatment schedule [data in Fig.6(b)] and intensity of laser radiation [Table 1 and Fig. 7(a)].

4. Summary

In summary, the melt–quenching approach was employed to prepare Au NP–embedded sodium zinc borate bearing the composition $29.9\text{Na}_2\text{O}–10\text{ZnO}–60\text{B}_2\text{O}_3–0.1\text{AuCl}_3$ (mol%). The efficacy of annealing duration on NLO traits of Au NP–embedded sodium zinc borate glass was investigated and superior NLO magnitudes could be observed for 12 hr of heat treatment at 450°C . The intensity–dependent NLO investigations on optimized NZBAu12 glass system found that the increased laser radiation input fluence led SA to switch to RSA nonlinearity. It was identified that under low intensity laser pulses, the intraband transitions and SPR bleaching was predominant, thereby resulting in SA type of nonlinear absorption. Whilst under high fluence laser pulses, the intraband transitions were subordinate to the additional interband and mixed intraband–interband transitions, which in the excited state produced free carriers and hence resulted the RSA nonlinearity due to 2PA. The NLO investigations clearly revealed the NZBAu12 glass's benefits for optical Q–switching functionalities under low–energy laser pulse, whilst the glass was found to be beneficial for optical limiting applications under high fluence laser pulses.

CRediT authorship contribution statement

K. Gurushantha: Conceptualization, Methodology, Data curation, Writing – original draft.

Harika Patnala: Data curation, Writing – review & editing. **Sangeeta B. Kolavekar:**

Conceptualization, Methodology, Writing – review & editing. **Vadiraj B. Tangod:** Data

curation, Writing – review & editing. **Dalal Abdullah Aloraini:** Data curation, Writing –

review & editing. **Aljawhara H. Almuqrin:** Data curation. **M.I. Sayyed:** Data curation. **S.**

Venugopal Rao: Validation, Writing – review & editing. **A.G. Pramod:** Methodology, Data

curation, Writing – original draft. **Gangareddy Jagannath:** Data curation, Validation,

Writing – review & editing. **D. Rajeshree Patwari:** Data curation, Writing – review &

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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...The O2-2 are unstable and are connected only loosely with network cations. The outer O2-2 electrons are screened from the nucleus during the high fluence light irradiation and result in anharmonic effects that provoke the glasses' optical non-linearity [51]. With the increased content of AgNO3 in the glass composition, the number density of Ag0 atoms enhance, which leads to increased Ag NP size that then enhances the NBO content and ultimately causes glasses with AgNO3 content to have enhanced NLO properties...

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